

Moist Carrot Cake

Macerated + pressed carrot. Toasted-walnut butter distributed through the crumb. Mascarpone-tempered cream cheese frosting that holds at galley temperature.

Yield: **12 portions** (9×13" sheet) Active: **25 min** Total: **1 hr 40 min** Oven: **175°C** (350°F) Done: **95°C** internal

WHERE IT WAS BORN Medieval Europe, where sweet roots stood in for costly sugar — an English “pudding in a Carret root” appears by 1591; revived in WWII Britain under the Ministry of Food’s <i>Dr Carrot</i> .	EATEN, AUTHENTICALLY The ration-table cake: naturally sweetened with carrot when sugar ran short, eaten plain or sugar-dusted as a teacake — no frosting in sight.	HOW IT’S USED NOW An American classic since the 1960s health-food revival; the cream-cheese frosting (Philadelphia, 1965) is its modern signature.
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INGREDIENTS

METRIC · US VOLUME IN RIGHT COLUMN

MACERATION

Carrots, peeled, grated	400g	~5 med
Light brown sugar (<i>for maceration</i>)	30g	2 tbsp
Fine sea salt (<i>draw moisture</i>)	3g	½ tsp

DRY

All-purpose flour	260g	2¼ cups
Baking soda	6g	1 tsp
Baking powder	5g	1¼ tsp
Cinnamon, ground	5g	2 tsp
Ginger, ground	3g	1 tsp
Nutmeg, fresh-grated	1g	¼ tsp
Diamond kosher salt	5g	1 tsp

WET

Light brown sugar	190g	scant 1 cup
Granulated sugar	100g	½ cup
Neutral oil (<i>grapeseed/sunflower</i>)	200g	scant 1 cup
Eggs, large (<i>RT</i>)	150g	×3
Pure vanilla extract	8g	1½ tsp
Toasted-walnut butter (<i>sub-recipe</i> ↓)	60g	3 tbsp
Orange zest, microplaned	4g	1 orange

FROSTING

Cream cheese, full-fat (<i>cold</i>)	340g	12 oz
Mascarpone (<i>cold</i>)	115g	½ cup
Unsalted butter, soft (<i>20°C</i>)	85g	6 tbsp
Icing sugar, sifted	280g	2¼ cups
Pure vanilla extract	5g	1 tsp

Kit: 23×33 cm (9×13") pan · food processor · colander + towel · scale · thermometer · stand mixer · offset spatula · parchment

METHOD

1. **Macerate carrot.** Toss 400g grated carrot with 30g brown sugar + 3g salt. Colander over bowl. Stand **20 min**.

Osmosis draws ~30% free water out of carrot cells → concentrated flavour, less soggy crumb. — McGee 2004

2. **Press dry.** Squeeze in clean towel. Target ≤280g pressed carrot. Reserve 40g liquid (optional glaze).

SUB-RECIPE — TOASTED WALNUT BUTTER
Raw walnut halves, 160g · pinch fine salt (optional) · *yields ~70g butter + 80g chopped garnish*

2a. Toast all 160g at **165°C (325°F)**, 10–12 min. Cool 5 min.

3a. Split: **80g → food processor** 3–4 min → smooth butter (~70g). Use 60g in batter; 10g extra keeps 2 wk fridge.

4c. **Remaining 80g:** rough chop for garnish.

5d. *Maillard furaneol + pyrazines in the fat phase → roasted note in every bite. — López-Alt / Modernist*

3. **Preheat 175°C** (350°F). Butter + parchment 9×13" pan. Whisk all dry together.

4. **Whisk wet.** 190g brown sugar, 100g granulated, 200g oil, 150g eggs, vanilla, 60g walnut butter, orange zest. 1 min smooth.

5. **Fold carrot into wet.** 4 folds.

6. **Add dry in 2 additions.** Half → 6 strokes → second half → fold until just incorporated. Flour streaks fine. Tap pan twice. Bake **28–34 min**.

Soda reacts with brown sugar acidity immediately; powder activates on heat. Sequential lift prevents collapse before crumb sets. — McGee 2004
- Done:** Internal **95°C** (203°F). Top springs back. Edges pull from pan. Toothpick: moist crumbs only.
Cool: 15 min in pan → wire rack. Frost only at ≤22°C (at least 60 min).
- Frosting** — mix while cake cools
- 6a. Paddle **cold cream cheese** 30 sec only. Add cold mascarpone, beat 20 sec.

7a. Add softened butter, beat 60 sec. Sift icing sugar in 2 additions, mix to incorporate only.

8c. Vanilla + pinch salt. Taste. If slack: refrigerate 15 min. *Mascarpone (60–75% fat) buffers frosting stability at 28°C. — ATK 2021*

9d. **Frost:** Offset spatula, 1 cm layer. Scatter toasted walnuts + orange zest. Chill 30 min before slicing.
- ### ONE BATTER, TWO DISHES — SPICED CARROT TEA-LOAF
- The frosting-free original. Pour the same batter into **two 900 g (2 lb) loaf tins**, two-thirds full; bake **175°C, 45–55 min** to 95°C internal (foil at 35 min if browning). No frosting — glaze with the reserved carrot liquid + sugar, or leave plain. *Keeps 2–3 days at room temp, no cold chain — the smart passage bake and the crew tier.*
- ## ELEVATION
- ### TIER 1 — NO EXTRA TIME
- Cardamom + white pepper** — 1g + 0.5g in dry
 - Miso in frosting** — 10g white miso with butter
 - Dark brown sugar swap** — more molasses depth
- ### TIER 2 — WORTH IT (+15 MIN)
- Carrot-liquid glaze** — reduce 40g liquid + 20g sugar, brush warm cake
 - Candied ginger** — 50g diced, fold in at end
 - Part brown butter** — replace 100g oil with 85g browned
- ### TIER 3 — RESTAURANT
- Dehydrated carrot chips** — mandoline + 65°C / 2 hr, press into frosting
 - Pineapple soak** — 40g cold juice, brush warm cake
 - Toasted coconut** — 40g golden, press into fresh frosting
- ## TROUBLESHOOT
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| Dense, wet crumb | Carrot not pressed. Target ≤280g. Squeeze in towel. |
| Frosting weeps | Cream cheese too warm or over-beaten. Start cold. 30 sec max on CC alone. |
| Sunk centre | Underbaked. Hit 95°C internal. Lower oven to 165°C if recurring. |
| Rubbery crumb | Overmixed. Count 12 strokes max after dry goes in. |
| Frosting too sweet | Add 10g cold cream cheese, mix 10 sec. Or pinch more salt. |
| Muddy spice | Stale spices. Replace if >6 months old. Freshly grate nutmeg always. |
- ## CHARTER PREP & STORAGE
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| Freeze unfrosted | Always. Cream cheese frosting breaks on thaw. |
| Naked cake | Cling + foil. 2 mo freezer. Thaw overnight in fridge. |
| Frosted cake | 3 d fridge. Cover loosely — no cling on frosting. |
| Walnut butter | 2 wk fridge, jar. Stir if oil separates. |
| Pressed carrot | 3 d fridge, airtight. Drain before using. |
| Scaling | ×2 = two pans, same time. 8" rounds: 30–35 min. Muffins: 20–24 min. |
- CONTACT THE CHEF @arnaudcallier · www.littoralicious.com
- Science: McGee · Parks (BraveTart) · López-Alt · Modernist Cuisine · America’s Test Kitchen · King Arthur Baking
- The Blueprint